they're related, the oceans being the sites of origin, and the oxygen the product, of abundant life.

When you look carefully at the infrared spectrum of the Earth, you discover the minor constituents of the air. In addition to water vapor, there's carbon dioxide (CO_2), methane (CH_4), and other gases that absorb the heat that the Earth tries to radiate away to space at night. These gases warm the planet. Without them, the Earth would everywhere be below the freezing point of water. You've discovered this world's greenhouse effect.

Methane and oxygen together in the same atmosphere is peculiar. The laws of chemistry are very clear: In an excess of O_2, CH_4 should be entirely converted into H_2O and CO_2. The process is so efficient that not a single molecule in all the Earth's atmosphere should be methane. Instead, you find that one out of every million molecules is methane, an immense discrepancy. What could it mean?

The only possible explanation is that methane is being injected into the Earth's atmosphere so quickly that its chemical reaction with O_2 can't keep pace. Where does all this methane come from? Maybe it seeps out of the deep interior of the Earth—but quantitatively this doesn't seem to work, and Mars and Venus don't have anything like this much methane. The only alternatives are biological, a conclusion that makes no assumptions about the chemistry of life, or what it looks like, but follows merely from how unstable methane is in an oxygen atmosphere. In fact, the methane arises from such sources as bacteria in bogs, the cultivation of rice, the burning of vegetation, natural gas from oil wells, and bovine flatulence. In an oxygen atmosphere, methane is a sign of life.

That the intimate intestinal activities of cows should be detectable from interplanetary space is a little disconcerting, especially when so much of what we hold dear is not. But an alien scientist flying by the Earth would, at this point, be unable to deduce bogs, rice, fire, oil, or cows. Just life.

All the signs of life that we've discussed so far are due to comparatively simple forms (the methane in the rumens of cows is generated by bacteria that homestead there). Had your spacecraft